A square S has source and target horizontal maps, left and right vertical maps, and evidence that the two composites agree. Its quotient map is the map between cokernels induced by the right vertical map. For an inclusion of topological pairs, the four corners are the singular chains on the relative and ambient subsets of the source and target pairs.

Result 1 (quotient square commutes). [Topology foundations, cokernel.map]
Let K be a type. Let *field* be a field structure on K. Let C, D, E and F be elements of Ch_K. Let S be a commuting square of chain maps. Then

$$\overline{S.right} \circ \pi_{S.\mathrm{source}} = \pi_{S.\mathrm{target}} \circ S.\mathrm{right}.$$

Adapted from the cited source.

Result 2 (pair inclusion induces a chain square). [Topology foundations, singular chain functor applied to pair inclusions]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be a certified inclusion between pairs in X. Then

$$(\text{there exists } S : \mathrm{ChainSquare},\ S \text{ is the chain square of } Q)\,.$$

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.